INCH-POUND
MIL-M-85522A(AS)
17 June 1991
SUPERSEDING
MIL-M-85522(AS)
5 August 1983

MILITARY SPECIFICATION

MONITOR, OXYGEN, CRU-91/A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the oxygen monitor which, as part of the on-board oxygen generation system (OBOGS), monitors the delivered breathing gas mixture and provides a warning signal when oxygen concentration falls below prescribed levels. The oxygen monitor uses a replaceable polarographic sensor to detect oxygen partial pressure.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2b).

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner.
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS (continued)

MILITARY

MIL-P-116	-	Preservation, Methods of.
DOD-D-1000	-	Drawings, Engineering and Associated Lists.
MIL-E-5400	-	Electronic Equipment, Aerospace, General Specification for.
MIL-O-27210	-	Oxygen, Aviator's Breathing, Liquid and Gas.
MIL-G-27617	-	Grease, Aircraft and Instrument, Fuel and Oxidizer Resistant.

STANDARDS

MILITARY

DOD-STD-100	-	Engineering Drawing Practices.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-210	-	Climatic Information To Determine Design and Test Requirements for Military Systems and Equipment.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility Requirements for Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-704	-	Aircraft Electric Power Characteristics.
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines.
MIL-STD-970	-	Standards and Specifications, Order of Precedence for the selection of.
MIL-STD-1359	-	Cleaning Methods and Procedures for Breathing Oxygen Equipment.
MIL-STD-1523	-	Age Controls of Age-Sensitive Elastomeric Material (for Aerospace Applications).
MIL-STD-2073-1		DOD Materiel Procedures for Development and Application of Packaging Requirements (Part 1 of 2 Parts). Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the soliciation.

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DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1439AS101 - Monitor, Oxygen.

(Copies of the above specifications, drawings, and publications are available from the contracting activity or may be obtained as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The oxygen monitors furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.3 and 6.4). In addition, the retention of the qualification for the oxygen monitors on the applicable qualified products list shall be dependent on periodic verification of continued compliance with the requirments of this specification (see 4.3 and 4.3.2).

3.2 First article. When specified, an oxygen monitor sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein or in the applicable detail specifications shall be selected in accordance with MIL-STD-970.

3.4 Materials. Materials shall conform to referenced specifications and shall be as specified herein and on referenced drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.4.1 Metal parts. All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. The requirements of MIL-STD-454, Requirement 16 for Class 2 equipment shall apply.

3.4.2 Nonmetallic materials. Any nonmetallic material that is adversely affected by continued use with oxygen shall not be used.

3.4.2.1 Age. Elastomer components shall be controlled in accordance with MIL-STD-1523 (see 6.3).

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3.4.2.2 Ozone resistance. Elastomer components shall be composed of an ozone resistant composition which shall not bloom.

3.4.2.3 Fungus-proof materials. All materials used in the monitor construction shall be incapable of supporting fungoid growth, as defined in MIL-STD-810.

3.4.2.4 Operating life. The oxygen monitor shall have a minimum operating life of 8,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be approved for use by the acquisition activity.

3.4.2.5 Storage life. The oxygen monitor shall be designed to meet the performance and service life requirements of this specification after 5 years storage under worldwide climatic conditions specified in MIL-STD-210, when packaged in accordance with MIL-STD-2073-1, and MIL-P-116, Method IIb.

3.4.2.6 Protective treatment. When materials are used in the construction of the oxygen monitor that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip or scale during normal service life or under extremes of environmental conditions shall not be used.

3.5 Design and construction. The design and construction of the oxygen monitor shall be in accordance with MIL-E-5400 and envelope Drawing 1439AS101. The oxygen monitor shall conform to MIL-STD-454, Requirement 9, for workmanship.

3.5.1 Gas supply. The oxygen monitor shall be capable of operating at the flow-rates and pressures specified in 3.6.1.1 and shall be capable of withstanding a static inlet pressure of 75 psig without damage.

3.5.2 Electrical power. The oxygen monitor shall operate from a DC power source, in accordance with MIL-STD-704, with a voltage range for operation of 18.0 to 31.5 volts DC. Total power consumption including heater/accessories operation, shall be not greater than 56 watts at 28 VDC. Maximum power, without heater/accessories operation, shall be not greater than 7 watts at 28 VDC.

3.5.3 Electronic modules and solid state design.

3.5.3.1 Modules. The electronic portions of the oxygen monitor shall be functionally modularized.

3.5.3.2 Solid-state design. No electron tubes shall be utilized in the design of the oxygen monitor. Semi-conductor transistors and diodes shall be in accordance with MIL-STD-454, Requirement 30.

3.5.4 Altitude compensation. The oxygen monitor shall be capable of controlling the absolute pressure in the sensor cavity of 211.6 ± 4 mm Hg (31,400 ± 400 feet altitude) at altitudes above 31,800 feet.

3.5.5 Calibration. If manual calibration is required, the oxygen monitor shall contain a calibration adjustment that is easily accessible without disassembly.

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3.5.6 Output signal.

3.5.6.1 Analog signal. The oxygen monitor shall provide an analog signal voltage output which is directly proportional to the oxygen partial pressure of the sample gas.

3.5.6.2 Warning signal. The monitor shall activate a warning signal with a sample gas at an oxygen partial pressure of 182 +3/-12 mm Hg. within the normal temperature range of 40°F and 150°F, 182 +3/-17 mm Hg within the high temperature range of 150°F and 160°F, and 182 ± 13 mm Hg within the low temperature range of -65°F and 40°F.

3.5.6.3 Warning signal "on/off" range. The hysteresis exhibited between the warning "on" level and the warning "off" level (that is, the point at which the warning light is activated "on" with decreasing oxygen partial pressure and "off" with increasing oxygen partial pressure) shall be equivalent to no more than a 20 mm Hg partial pressure differential in oxygen concentration.

3.5.7 Press-to-test. The oxygen monitor shall contain a press-to-test feature capable of operating as specified in 3.6.1.2.

3.5.8 Leakage. With 5 psig applied to the oxygen monitor and all venting ports sealed, outward leakage shall be not greater than 0.010 lpm (NTP).

3.5.9 Fluid plumbing and electrical interface. The sample gas plumbing port and the electrical connector shall be as defined and located in Drawing 1439AS101.

3.5.10 Weight. The total weight of the oxygen monitor shall be not greater than 1.75 pounds.

3.5.11 Interchangeability. The oxygen monitor shall meet the interchangeability requirements of MIL-E-5400, which includes MIL-STD-454, Requirement 7. In addition, all parts, subassemblies and assemblies having the same part number shall be identical in accordance with DOD-STD-480 and DOD-STD-100. When any part is interchanged, the oxygen monitor shall meet all performance limits with calibration adjustments only.

3.5.12 Identification of product. The oxygen monitors shall be identified in accordance with MIL-STD-130.

3.5.13 Serial numbers. The oxygen monitors shall be identified by individual serial numbers which shall be assigned by the manufacturer. Serialization shall be by a block of consecutive numbers to cover the entire acquisition quantity.

3.5.14 Cleanliness.

3.5.14.1 Degreasing. Prior to assembling the oxygen monitor and its subassemblies, all internal surfaces shall be degreased in accordance with MIL-STD-1359. After completion of the cleaning and when assembled, a Halide testing apparatus having a sensitivity of at least 3 x 10^{-4} standard cc/sec, shall be used to determine the absence of the cleaning compound. When other than a halide containing compound is used to degrease, appropriate procedures shall be employed to assure complete cleaning compound removal.

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3.5.14.2 Lubricants. Lubricants qualified to MIL-G-27617 may be used sparingly on seals and straight threads if assembly difficulty is encountered.

3.6 Performance.

3.6.1 Operational.

3.6.1.1 Sample gas flow rate. The gas consumption of the oxygen monitor, when tested as specified in 4.8.2.1, shall be not greater than 0.20 lpm (NTP) with 4 psig input pressure and 2.0 Liters per minute (lpm) (NTP) with 72 psig input pressure.

3.6.1.2 Press-to-test. The oxygen monitor, when tested as specified in 4.8.2.2, shall activate the warning light at the oxygen partial pressure specified in 3.6.5.2 within 25 seconds in both the normal temperature range of 40°F to 150°F and the high temperature range of 150°F to 160°F; and within 60 seconds in the low temperature range of -65°F to 40°F. The warning signal shall be deactivated after release of the press-to-test plunger within 15 seconds in both the normal range of 40°F to 150°F and the high temperature range of 150°F to 160°F; and within 20 seconds in the low temperature range of -65°F to 40°F.

3.6.1.3 Altitude. The oxygen monitor, when tested as specified in 4.8.2.3, shall activate the warning light at an oxygen partial pressure of 182 +3/-12 mm Hg.

3.6.2 Voltage variation. The oxygen monitor, when tested as specified in 4.8.3, shall activate the warning signal at an oxygen partial pressure of 182 +3/-12 mm Hg. Total current draw, to include warning and heater activation, shall not exceed 3.5 amperes.

3.6.3 Electromagnetic interference/electromagnetic compatibility. The oxygen monitor, when tested as specified in 4.8.4, shall show no evidence of functional, electrical or material failure during or after emission and susceptibility testing, and shall be not greater than the limits specified for Class A1 equipment, Category A1b.

3.6.4 High temperature. The oxygen monitor, when tested as specified in 4.8.5, shall show no evidence of functional, electrical or material failure and the warning circuit shall be activated at the oxygen partial pressure specified in 3.5.6.2.

3.6.5 Low temperature.

3.6.5.1 Operational. The oxygen monitor, when tested as specified in 4.8.6.1, shall show no evidence of functional, electrical or material failure. The warning signal shall be activated at the oxygen partial pressure specified in 3.5.6.2..

3.6.5.2 Warm up. The oxygen monitor, when tested as specified in 4.8.6.2, shall attain the specified analog signal within 15 minutes at both -65°F and -15°F. The time period, at which the warning signal remains deactivated, shall not exceed 10 minutes at -15°F and 12 minutes at -65°F.

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3.6.6 Temperature/altitude. The oxygen monitor, when tested as specified in 4.8.7 and 4.8.7.1, shall show no evidence of functional, electrical or material failure. The warning signal shall be activated at the oxygen partial pressure specified in 3.5.6.2.

3.6.7 Temperature shock. The oxygen monitor, when tested as specified in 4.8.8, shall show no evidence of functional, electrical or material failure following the temperature shock exposure.

3.6.8 Humidity. The oxygen monitor, when tested as specified in 4.8.9, shall show no evidence of functional, electrical or material failure following the humidity exposure.

3.6.9 Temperature/humidity/altitude. The oxygen monitor, when tested as specified in 4.8.7.2, shall show no evidence of functional, electrical or material failure following the environmental exposure.

3.6.10 Acceleration. The oxygen monitor, when tested as specified in 4.8.10, shall show no evidence of functional, electrical or material failure during or following the acceleration exposure. The analog output during operational level acceleration shall not deviate more than ± 10 mm Hg.

3.6.11 Vibration. The oxygen monitor, when tested as specified in 4.8.11, shall show no evidence of functional, electrical or material failure during or following the vibration exposure. The analog output of the monitor during operational level vibration shall not deviate more than ± 10 mm Hg.

3.6.12 Shock. The oxygen monitor, when tested as specified in 4.8.12, shall show no evidence of functional, electrical or material failure following shock exposure.

3.6.13 Salt fog. The oxygen monitor, when tested as specified in 4.8.13, shall show no evidence of functional, electrical or material failure following salt fog exposure.

3.6.14 Rain. The oxygen monitor, when tested as specified in 4.8.14, shall show no evidence of functional, material or electrical failure following rain exposure.

3.6.15 Dust. The oxygen monitor, when tested as specified in 4.8.15, shall show no evidence of functional, electrical or material failure following dust exposure.

3.7 Workmanship. The oxygen monitors shall be uniform in quality and shall be free from irregularities, defects, or foreign material which could adversely affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

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disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection. Qualification inspection shall consist of the examinations and tests performed on samples submitted for approval as a qualified product (see 4.3).

b. First article inspection. First article inspection shall consist of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 4.4).

c. Quality conformance inspection. Quality conformance inspection shall consist of exmaninations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements specified in this specification (see 4.5).

d. Quality conformance verification inspection. Quality conformance verification inspection shall consist of examinations and tests performed on individual products or lots to verify conformance of the products or lots with requirements specified in this specification (see 4.6).

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in Table I.

4.3.1 Samples. Qualification samples shall consist of six complete oxygen monitors and one set of manufacturer's drawings. Samples shall be forwarded to a test facility specified in the letter of authorization to submit samples (see 6.4). The samples shall be plainly identified by securely attached durable tags marked with the following information:

Sample submitted by (name), (date) for qualification inspection in accordance with the requirements of MIL-M-85522A(AS) and number under authorization (reference authorizing letter and number) (see 6.4).

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4.3.2 Retention of qualification. The retention of qualification shall consist of verification to determine compliance of the qualified oxygen monitor with the requirements of this specification. Every two years, each manufacturer will be requested to forward to the Government certification signed by a responsible official of management, attesting that the listed product(s) is still available from the listed plant, can be produced under the same conditions as originally qualified; i.e., same process, materials, construction, design, manufacturer's part number, or designation; and meets the requirements of the current issue of the specification. Failure to provide the certification will be cause for removal from the Qualified Products List (QPL). After completion of the certification review, the QPL shall be reprinted to show the date of validation.

4.4 First article inspection. First article inspection shall consist of the examinations and tests specified in Table II.

4.4.1 Samples. Unless otherwise specified in the contract or order, as soon as practicable after the award of the contract or order, the manufacturer shall submit a minimum of two complete oxygen monitors. The samples shall be representative of the construction, workmanship, components and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article samples on a new contract or whether first article reliability test is required on a new contract may be waived at the discretion of the acquiring activity (see 6.2.f). Approval of the first article samples or the waiving of the first article inspection does not waive the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as specified in the contract.

4.4.1.1 Documentation. Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production shall be made available to the cognizant Government activity (see 6.2.f and 6.3). The monitors will be returned to the manufacturer for use in monitoring production.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in Table III. The sampling and inspection levels and acceptance criteria shall be as specified in MIL-STD-105.

4.5.1 Sampling.

4.5.1.1 Inspection lot.

4.5.1.1.1 Monitors. An inspection lot size shall be expressed in units of one oxygen monitor made under essentially the same conditions and from the same materials and components. The sample unit shall be one oxygen monitor.

4.5.1.1.2 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing oxygen monitors, fully prepared for delivery, made from the same materials and components. The sample unit shall be one shipping container, containing oxygen monitors, fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of monitors. The sample size, acceptance criteria, tests and examinations required for the oxygen monitors shall be as specified in Table III.

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4.6 Quality conformance verification inspection. At the option of the Government and upon completion of the quality conformance inspection requirements of 4.5, a random sample, one for every 50 or fraction thereof, shall be selected from each lot of oxygen monitors. Each oxygen monitor, selected as a sample unit, shall be forwarded to a laboratory designated at the time of award (see 6.2.g). The designated laboratory shall conduct the following tests and examinations:

a. Visual examination
b. Operational test
c. Voltage variation
d. High temperature
e. Low temperature

The tests may be conducted in any order. The serial numbers of the units in the lot, represented by the sample units, shall be included with the data accompanying the samples to the laboratory. The Government activity responsible for conducting the quality conformance verification program (see 6.2.g) shall report the results of the tests and examinations to the designated inspection and acceptance activity specified in the acquisition document. Final acceptance of the lot from which the sample units were selected shall be based upon successful completion of the inspection program by the cognizant Quality Assurance Representative/Specialist at the contractor's facility; applying the applicable acceptance criteria specified in Table III.

4.7 Test conditions. Unless otherwise specified in the inspection method, tests shall be conducted at local ambient temperatures and barometric pressure. Corrections shall be made to provide agreement with the temperature and pressure calibration of the instruments. Inspection data provided by any instrument not calibrated to normal temperature and pressure (NTP) conditions shall be corrected to determine NTP requirements. NTP conditions are 29.92 inches of mercury (101.3kPa) and 70°F (21.1°C).

4.7.1 Gas. Unless otherwise specified herein, the gas used in testing the oxygen monitors shall be oxygen in accordance with MIL-O-27210, Type I.

4.8 Inspection methods.

4.8.1 Visual examination.

4.8.1.1 Oxygen monitors. Every oxygen monitor shall be examined visually for critical defects to determine conformance to this specification. In addition, every oxygen monitor, selected as a sample unit from the lot, shall be visually examined for minor defects to determine conformance to this specification. The classification of defects specified in Table IV shall be used to classify the defects found.

4.8.1.1.1 Dimensions. Each oxygen monitor, selected as a sample unit from the lot, shall be checked dimensionally to determine conformance to the dimensions in Drawing 1439AS101.

4.8.1.2 Packaging. Each of the fully prepared shipping containers, containing oxygen monitors, selected as a sample unit from the lot, shall be examined to determine that the packaging, packing and marking conform to this specification. The list of defects specified in Table V shall be used to enumerate the defects found.

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4.8.2 Operational tests.

4.8.2.1 Sample gas flow rate. The oxygen monitor shall be connected to a source of 28 VDC electrical power and the sample gas inlet connected to an oxygen source capable of delivering 4 through 72 psig. With an inlet pressure of 4 psig, the sample gas flow rate shall be recorded. The test shall be repeated with an inlet pressure of 72 psig. The oxygen monitor shall pass the requirements specified in 3.6.1.1.

4.8.2.2 Press-to-test. With the oxygen monitor operating with a supply pressure of 7 ± 1 psig, the press-to-test plunger shall be depressed. The analog output at warning and time to warning activation shall be recorded. With the plunger released, the analog output at warning deactivation and time to deactivation shall be recorded. The oxygen monitor shall pass the requirements specified in 3.6.1.2.

4.8.2.3 Altitude test. The oxygen monitor shall be operated with a sample gas inlet pressure of 25 psig and subjected to ambient altitudes of 31,800 and 50,000 feet. The oxygen concentration of the sample gas shall be reduced until a warning activation occurs. The analog output at warning shall be recorded and shall pass the requirements specified in 3.6.1.3.

4.8.3 Voltage variation test. The oxygen monitor shall be connected to a power source of variable voltage and oxygen with an inlet pressure of 25 psig. The press-to-test check of 4.8.2.2 shall be conducted with input voltages of 18.0 and 31.5 VDC. The oxygen monitor shall pass the requirements specified in 3.6.2.

4.8.4 Electromagnetic interference/electromagnetic compatibility. The oxygen monitor shall be tested in accordance with MIL-STD-461 and MIL-STD-462. During operation, the oxygen monitor shall be supplied with oxygen at 25 psig. The oxygen monitor shall pass the requirements specified in 3.6.3.

4.8.5 High temperature test. Unless otherwise specified herein, the oxygen monitor shall be tested in accordance with the high temperature test of MIL-STD-810, Method 501.3, Procedure II. The oxygen monitor shall be conditioned at 160°F ± 2°F for 8 hours. During conditioning, the monitor shall be operational with a source of 28 ± 2 VDC electrical power and the inlet connected to an oxygen source capable of delivering 4 to 72 psig. After conditioning, and while still at 160° ± 2°F, the oxygen monitor shall be subjected to the press-to-test performance check of 4.8.2.2. Upon completion of the press to test performance test, and while still at 160° ± 2°F, the partial pressure of oxygen delivered to the monitor shall be lowered until warning signal activation occurs. The partial pressure at activation shall be recorded. The monitor shall be returned to standard ambient conditions and subjected to the operational tests of 4.8.2. The monitor shall pass the requirements specified in 3.6.4.

4.8.6 Low temperature tests.

4.8.6.1 Operational. The oxygen monitor shall be operated with 28 ± 2 VDC and with an oxygen inlet pressure of 25 psig. Ambient temperature shall be lowered to -65° ± 2°F and maintained for a minimum period of 8 hours. After conditioning, and while still at -65° ± 2°F, the monitor shall be subjected to the press-to-test performance check of 4.8.2.2. Upon completion of the press-to-test performance check and while still at -65° ± 2°F, the partial

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pressure of oxygen supplied to the oxygen monitor shall be lowered until warning light activation occurs. Record the partial pressure at activation. The monitor shall be returned to standard ambient conditions subjected to the operational tests of 4.8.2. The monitor shall pass the requirements specified in 3.6.5.1.

4.8.6.2 Warm up. The oxygen monitor shall be exposed to an ambient temperature of -65° ± 2°F for a period of 8 hours non-operating. After the 8 hour stabilization, the oxygen monitor shall be supplied with 28 ± 2 VDC and 40 percent oxygen at 25 psig. Any warning activation/deactivation during warm up shall be observed. The time it takes to deactivate the warning signal and remain deactivated after initial application of power and sample gas shall be recorded. In addition, analog input shall be monitored continuously until a value within ± 10 mm Hg at standard temperature and 40 percent oxygen is attained and stabilized. The time to reach this value from initial application of power and sample gas shall be recorded. After non-operational stabilization at room temperature, the test shall be repeated with an ambient temperature of -15° ± 2°F. The oxygen monitor shall pass the requirements specified in 3.6.5.2.

4.8.7 Temperature/humidity/vibration/altitude/ test. Unless otherwise specified herein, the oxygen monitor shall be subjected to the temperature/ humidity/vibration/ altitude test in accordance with MIL-STD-810, Method 520.1, Procedure III. This procedure shall be conducted as a series of two combination environmental tests:

a. Temperature/altitude (4.8.7.1)

b. Temperature/humidity/altitude (4.8.7.2).

4.8.7.1 Temperature/altitude. During temperature/altitude exposure times, the monitor shall be supplied with power and oxygen pressure extremes of 18 to 31.5 VDC and 4 to 72 psig, respectively. The oxygen monitor shall be subjected to the press-to-test performance check of 4.8.2.2, as required during each temperature/altitude condition. Following this environmental exposure, the monitor shall be subjected to the operational tests of 4.8.2 and shall pass the requirements of 3.6.6.

4.8.7.2 Temperature/humidity/altitude. During temperature/humidity/ altitude exposure times, the oxygen monitor shall be non-operational and the inlets shall be capped. Following the environmental exposure, the oxygen monitor shall be subjected to the operational tests of 4.8.2 and shall pass the requirements of 3.6.9.

4.8.8 Temperature shock test. Unless otherwise specified herein , the oxygen monitor shall be subjected to the temperature shock test in accordance with MIL-STD-810, Method 503.3. The high and low temperatures shall be 160° ± 2°F and -70° ± 2°F, respectively. During conditioning, the oxygen monitor shall be non-operational and the chamber temperatures shall be maintained for a period of not less than 4 hours. The exposure sequence shall consist of high temperature followed by low temperature. This sequence shall be repeated for a total of 3 cycles, After stabilization at room temperature, the oxygen monitor shall be subjected to the operational tests of 4.8.2, and shall pass the requirements specified in 3.6.7.

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4.8.9 Humidity test. Unless otherwise specified herein, the oxygen monitor shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507.3,. The monitor shall be non-operational and the inlet port shall be capped. Prior to conditioning, ambient temperatures shall be maintained at 129° ± 2°F for 24 hours to dry the monitor. The monitor shall be conditioned at 73° ± 2°F and 50 ± 10 percent relative humidity for 24 hours. Following conditioning, the chamber temperature shall be raised to 80° ± 2°F and the relative humidity shall be raised to 94 ± 4 percent. The monitor shall be subjected to five continuous cycles in accordance with Figure 5. Upon completion of cycling, the monitor shall be returned to standard ambient conditions and subjected to operational tests of 4.8.2, and shall pass the requirements specified in 3.6.8.

4.8.10 Acceleration test. Unless otherwise specified herein, the oxygen monitor shall be subjected to the acceleration test in accordance with MIL-STD-810, Method 513.4, Procedures I and II. During Procedure I testing, the oxygen monitor shall be non-operational, with a test level of 27 G in all axes. During Procedure II testing, the oxygen monitor shall be operated with 28 VDC and sample gas inlet pressures at 25 psig, with a test level of 18 G in all axes. Analog output shall be monitored continuously during each operational test. At the completion of each procedure, the oxygen monitor shall be subjected to the operational tests of 4.8.2. The oxygen monitor shall pass the requirements specified in 3.6.10.

4.8.11 Vibration test. Unless otherwise specified herein, the oxygen monitor shall be subjected to the vibration test in accordance with MIL-STD-810, Method 514.4, Procedure I, category 5 jet aircraft. Operational level testing shall be conducted with a power spectral density of 0.04 g^2/Hz, with a test time of one hour in each of the three mutually perpendicular axes. During operation, the oxygen monitor shall be supplied with 28 VDC and oxygen at 25 psig. Analog output shall be monitored continuously. Endurance level testing shall be conducted with a power spectral density of 0.2033 g^2/Hz, with a test time of one hour in each axis. The oxygen monitor shall be subjected to the operational tests of 4.8.2 after operational and endurance testing in each axis. The oxygen monitor shall pass the requirements specified in 3.6.11.

4.8.12 Shock test. The oxygen monitor shall be subjected to the shock test in accordance with MIL-STD-810, Method 516.4, Procedure I, Functional Shock, 20 g's peak and 11 ms nominal duration and Procedure IV, Transit Drop, Table 514.4-II, for under 100 pound category. The oxygen monitor shall be non-operational during both procedures. Following completion of each procedure, the oxygen monitor shall be subjected to the operational tests of 4.8.2 and shall pass the requirements specified in 3.6.12.

4.8.13 Salt fog test. Unless otherwise specified herein, the oxygen monitor shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509.3. The salt fog solution shall be as specified in Table VI. The test shall consist of a 48 hour exposure period followed by a 48 hour drying period. During exposure, the oxygen monitor shall be non-operational and the inlet port shall be capped. Immediately following exposure, the monitor shall be subjected to the operational tests of 4.8.2 followed by a visual inspection. If necessary to aid in the visual inspection, a gentle wash in running water not warmer than 100°F may be used. Following drying, the operational tests and visual inspection shall be repeated. The monitor shall pass the requirements in 3.6.13.

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4.8.14 Rain test. Unless otherwise specified herein, the oxygen monitor shall be subjected to the rain test in accordance with MIL-STD-810, Method 506.3, Procedure II. The temperature of the water shall be between 50°F and 95°F and the rainfall shall be produced at a rate of 2 ± 0.5 inches per hour. During exposure, the oxygen monitor shall be non-operational and the inlet port shall be capped. The monitor shall be positioned to allow impingement of the rain on the monitor normal to its horizontal plane for 1 hour minimum. Upon completion of normal exposure, the monitor shall be repositioned 90° from the horizontal in all four directions for 15 minutes, minimum, to allow exposure to rainfall from all significant surfaces. Following exposure, the monitor shall be subjected to a visual inspection. In order to aid the visual inspection, the monitor may be cleaned by sponging or lightly wiping accumulated moisture. Upon completion of the visual inspection, the monitor shall be subjected to the operational tests of 4.8.2. The oxygen monitor shall pass the requirements of 3.6.14.

4.8.15 Dust test. Unless otherwise specified, the oxygen monitor shall be subjected to the dust test in accordance with MIL-STD-810, Method 510.3, Procedure I. The test shall consist of three exposures to dust environments. During exposure times, the oxygen monitor shall be non-operational and the inlet port shall be capped. Exposure conditions are as follows:

Exposure	Temperature °F	Air Velocity ft/min	Dust Conc. gm/cuft	Time Hrs. (min)
1	73°F ± 2°F	1,750 ± 250	0.3 ± 0.2	6
2	145°F ± 2°F	300 ± 200	0.0	12
3	73°F ± 2°F	1,750 ± 250	0.3 ± 0.2	6

Following the third exposure, the oxygen monitor shall be subjected to the performance tests of 4.8.2 and shall pass the requirements specified in 3.6.15.

5. PACKAGING

5.1 Preservation. Unless otherwise specified in the contract or order, preservation shall be Level A, using Method IIb of MIL-P-116, with one oxygen monitor per unit package (see 6.2.h). Preservatives shall not be used. All openings in the oxygen monitor assembly shall be closed with suitable closures to prevent entry of dirt or foreign matter during shipment and storage.

5.2 Packing. Packing shall be Level A, B or C, as specified in the contract or order (see 6.2.h).

5.2.1 Level A. Each unit, packaged as specified in 5.1, shall be packed for shipment in containers conforming to PPP-B-601 or PPP-B-621, Overseas Type.

5.2.2 Level B. Each unit, packaged as specified in 5.1, shall be packed for shipment in containers conforming to PPP-B-601 or PPP-B-621, Domestic Type; fiberboard boxes conforming to PPP-B-636 or PPP-B-640, Weather Resistant may be used provided that the gross weight is not greater than the applicable specification limitation.

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5.2.3 Level C. Units packaged as specified in 5.1 shall be packed for shipment in exterior-type shipping containers conforming to rules and regulations pertaining to mode of transportation and acceptable to carrier at the lowest possible tariff rate that will ensure safe delivery. Contractor's commercial/industrial packing procedure may be used when the above criteria are met.

5.3 Marking. The exterior and interior containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall be required on each unit package and shipping container:

CAUTION

DO NOT ALLOW PETROLEUM CONTAMINANTS OF ANY KIND TO BE USED/STORED ON OR ABOUT THESE CONTAINERS.

5.4 Special requirements. All wrappings, cushioning, dunnage and containers used in preservation, packaging and packing of oxygen monitors shall be completely free of contamination by oil or grease.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The oxygen monitor covered by this specification is intended to be used in military aircraft breathing systems for monitoring of oxygen enriched air delivered by on-board oxygen generation systems and provide a warning signal for low oxygen level.

6.2. Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification including any amendment.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Government part number.

d. Quantity desired.

e. Whether first article inspection is waived (see 4.4.1).

f. Name and address of the first article inspection laboratory (see 4.4.1) and the name of the Government activity responsible for conducting the first article inspection program (see 4.4.1.1).

g. If required, name and address of the quality conformance verification inspection laboratory (see 4.6) and the name of the Government activity responsible for conducting the quality conformance verification inspection program (see 4.6).

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h. Applicable levels of preservation, packaging, and packing (see 5.1 and 5.2); including marking requirements (see 5.3) and special requirements (see 5.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirement List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

	Reference Paragraph	DID Number	DID Title	Suggested Tailoring
(a)	3.4.2.1	DI-E-2121	Certificate of compliance for the age of elastomer components	Use Contractor Format
(b)	4.4.1.1	DI-T-4902	First Article Inspection Report	Use contractor format

The above DID's were those cleared as of the date of this specification. The current use of DOD 5010.12-L Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL-85522) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Commander, Naval Air Systems Command, Washington, DC 20361; however, information pertaining to qualification of products may be obtained from the Naval Air Development Center (Code 6031), Warminster, PA 18974-5000. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031), indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.4.1 Documentation. When requested, the manufacturer shall make available engineering drawings and inspection reports in accordance with SD-6.

6.5 First article. When a first article inspection is required, the item(s) should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article

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inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Laboratory information. The successful bidder will be furnished with the name of the quality conformance verification inspection laboratory and the Government activity responsible for conducting the inspection program at the time of award. Samples from a rejected lot shall not be resubmitted for tests and examinations, as required by 4.6, without the approval of the contracting officer.

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to extensiveness of the changes.

6.8 Subject term (keyword) listing.
oxygen monitor
Gas contaminant
Sea level operation
Altitude operation
Oxygen generation system
Breathing gas

Preparing activity:
Navy - AS

(Project 1660-N591)

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TABLE I. Qualification inspection.

Inspection	Paragraph	
	Requirement	Inspection method
Visual examination	-	4.8.1
Operational tests	3.6.1	4.8.2
Voltage variation	3.6.2	4.8.3
EMI/EMC	3.6.3	4.8.4
High temperature	3.6.4	4.8.5
Low temperature	3.6.5	4.8.6
Temperature/altitude	3.6.6	4.8.7.1
Temperature shock	3.6.7	4.8.8
Humidity	3.6.8	4.8.9
Temperature/humidity/altitude	3.6.9	4.8.7.2
Acceleration	3.6.10	4.8.11
Vibration	3.6.11	4.8.12
Shock	3.6.12	4.8.13
Salt fog	3.6.13	4.8.14
Rain	3.6.14	4.8.15
Dust	3.6.15	4.8.16

TABLE II. First article inspection.

Inspection	Paragraph	
	Requirement	Inspection method
Visual examination	-	4.8.1
Operational tests	3.6.1	4.8.2
Voltage variation	3.6.2	4.8.3
High temperature	3.6.4	4.8.5
Low temperature	3.6.5	4.8.6

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TABLE III. Quality conformance inspection.

Inspection	Method	Sample size	Acceptance criteria
Visual examination (see classification of defects)	4.8.1.1	Every oxygen monitor for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defect. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensions	4.8.1.1.1	Inspection Level S-1 1/	Acceptance number zero, rejection number 1.
Operational	4.8.2	Every monitor	Reject all defective units.
Voltage variation	4.8.3	Every monitor	Reject all defective units.
High temperature	4.8.5	Inspection Level S-1 1/	Acceptance number zero, rejection number 1.
Low temperature	4.8.6	Inspection Level S-1 1/	Acceptance number zero, rejection number 1.
Reliability	4.8.17	Every oxygen monitor	Reject all defective units.
Packaging	4.8.1.2	Inspection Level S-2	Total acceptable quality level of 4.0 percent defective.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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TABLE IV. Classification of defects for visual examination of the oxygen monitor.

Critical	Minor
1. Material imperfections - foreign matter embedded.	201. Marking - missing, insufficient, incorrect, illegible or not permanent.
2. Surface - unclean, rough, misaligned, or containing cracks, nicks or other flaws.	202. Color not as specified.
3. Any component missing, malformed, fractured or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

TABLE V. List of defects for packaging.

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible, of improper size, location, sequence or method of application; markings not the same on the interior and exterior containers.
Packaging and packing materials	Any non-conforming component; any component missing, damaged or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, bulging or distorition of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements.

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TABLE VI. Salt fog solution.

Salt	Grams per liter
Sodium chloride	24.540
Magnesium chloride	11.110
Sodium sulfate	4.094
Calcium chloride	1.159
Potassium chloride	0.695
Sodium bicarbonate	0.201
Potassium bromide	0.101
Strontium chloride	0.042
Boric acid	0.027
Sodium fluoride	0.003

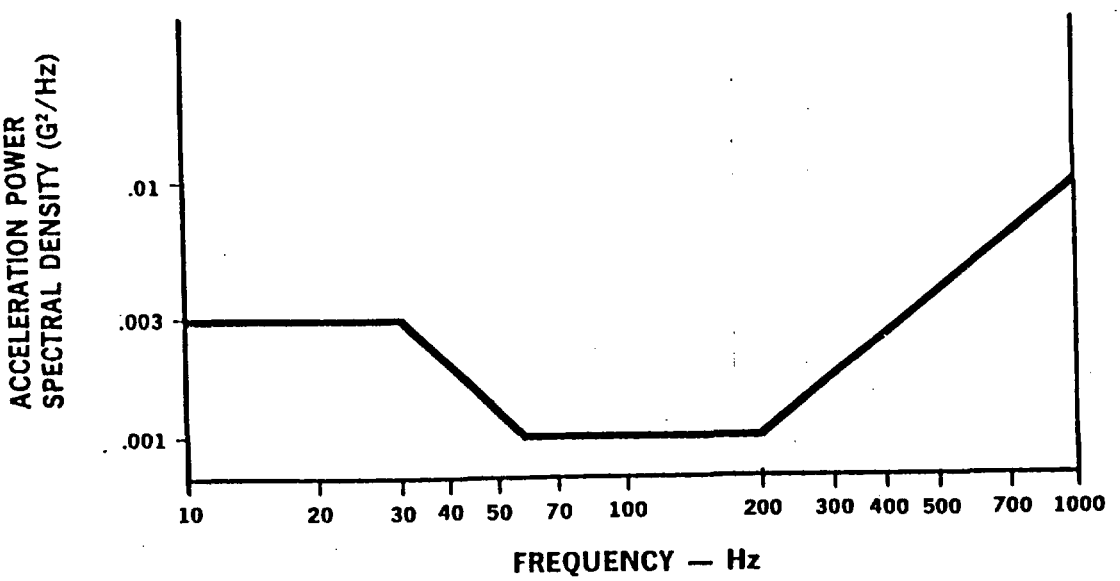
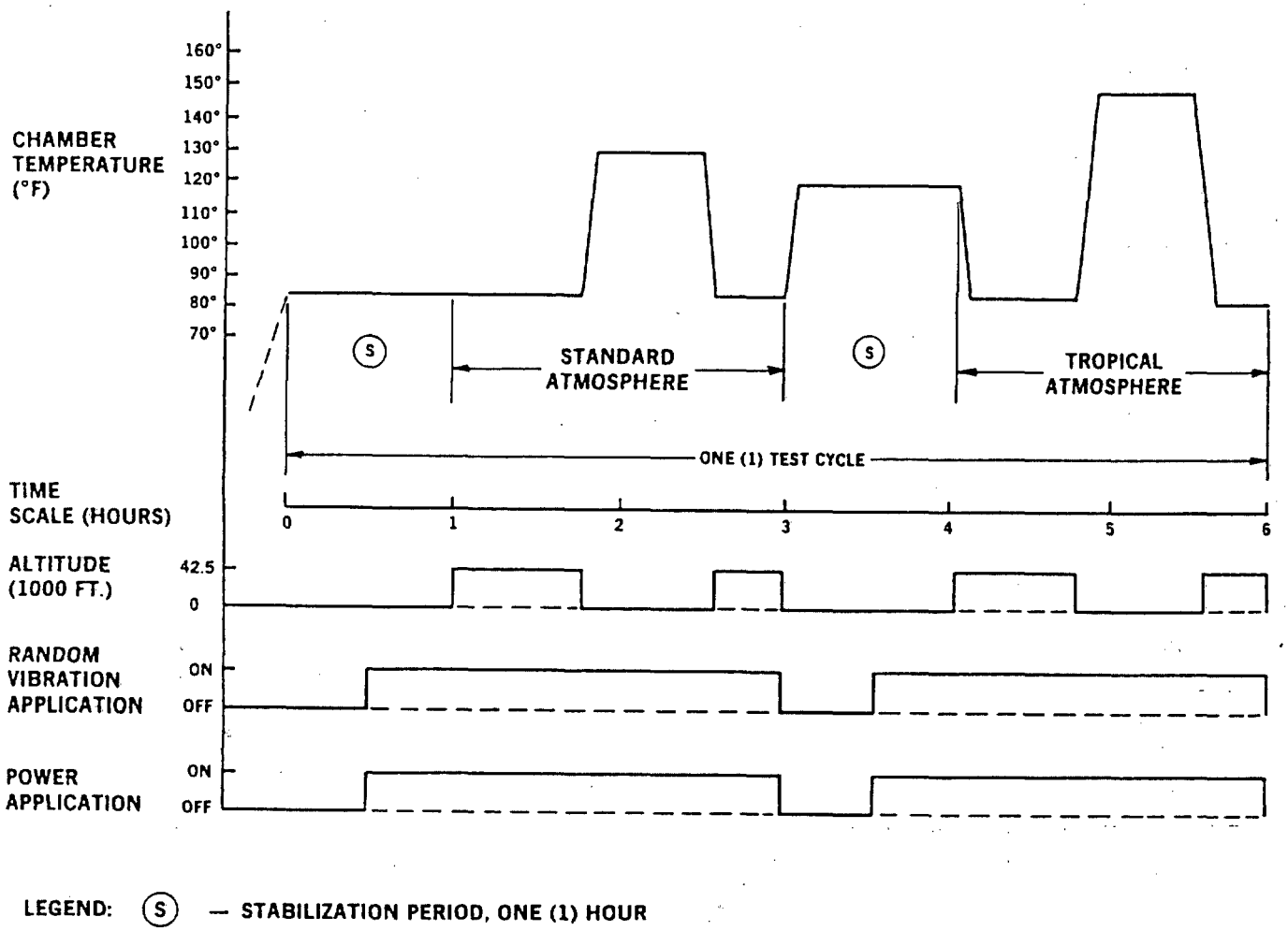


FIGURE 1. Random vibration envelope (reliability demonstrations).

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LEGEND: (S) — STABILIZATION PERIOD, ONE (1) HOUR

FIGURE 2. Oxygen monitor reliability test cycle.

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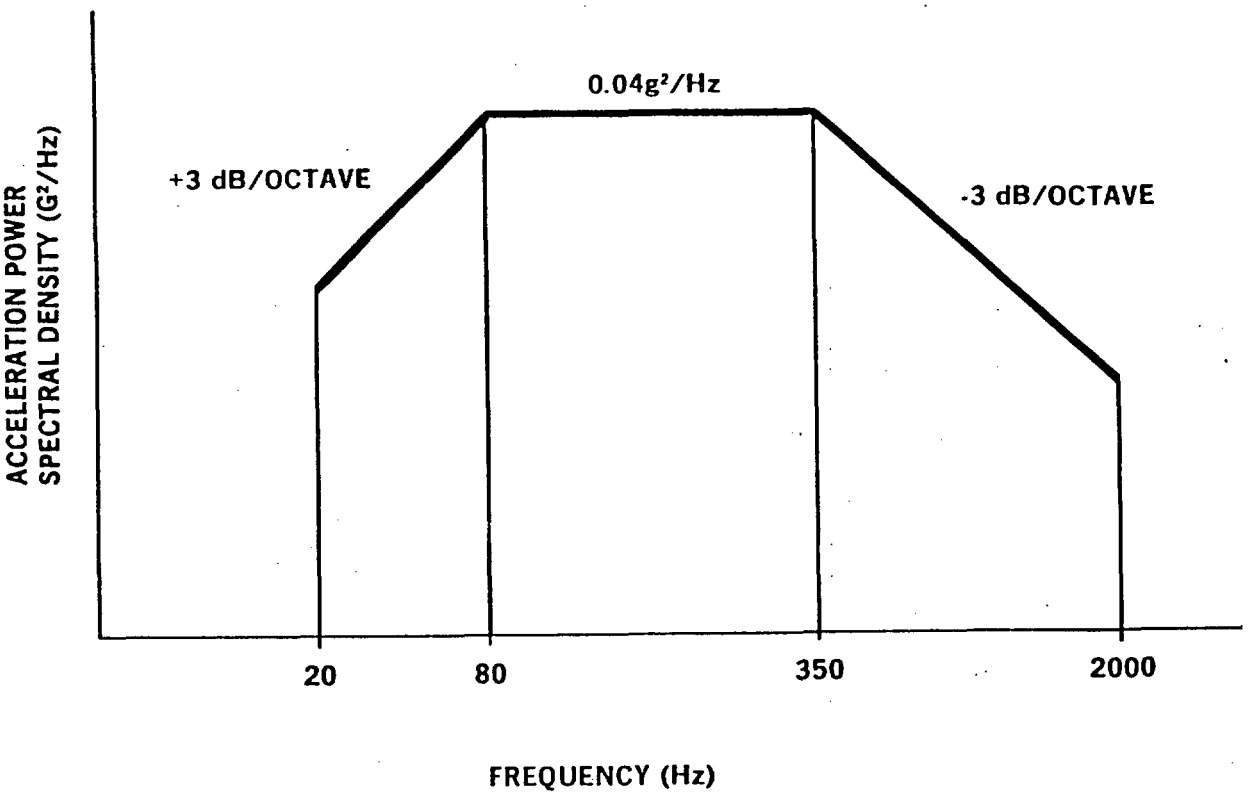


FIGURE 3. Random vibration envelope (reliability acceptance).

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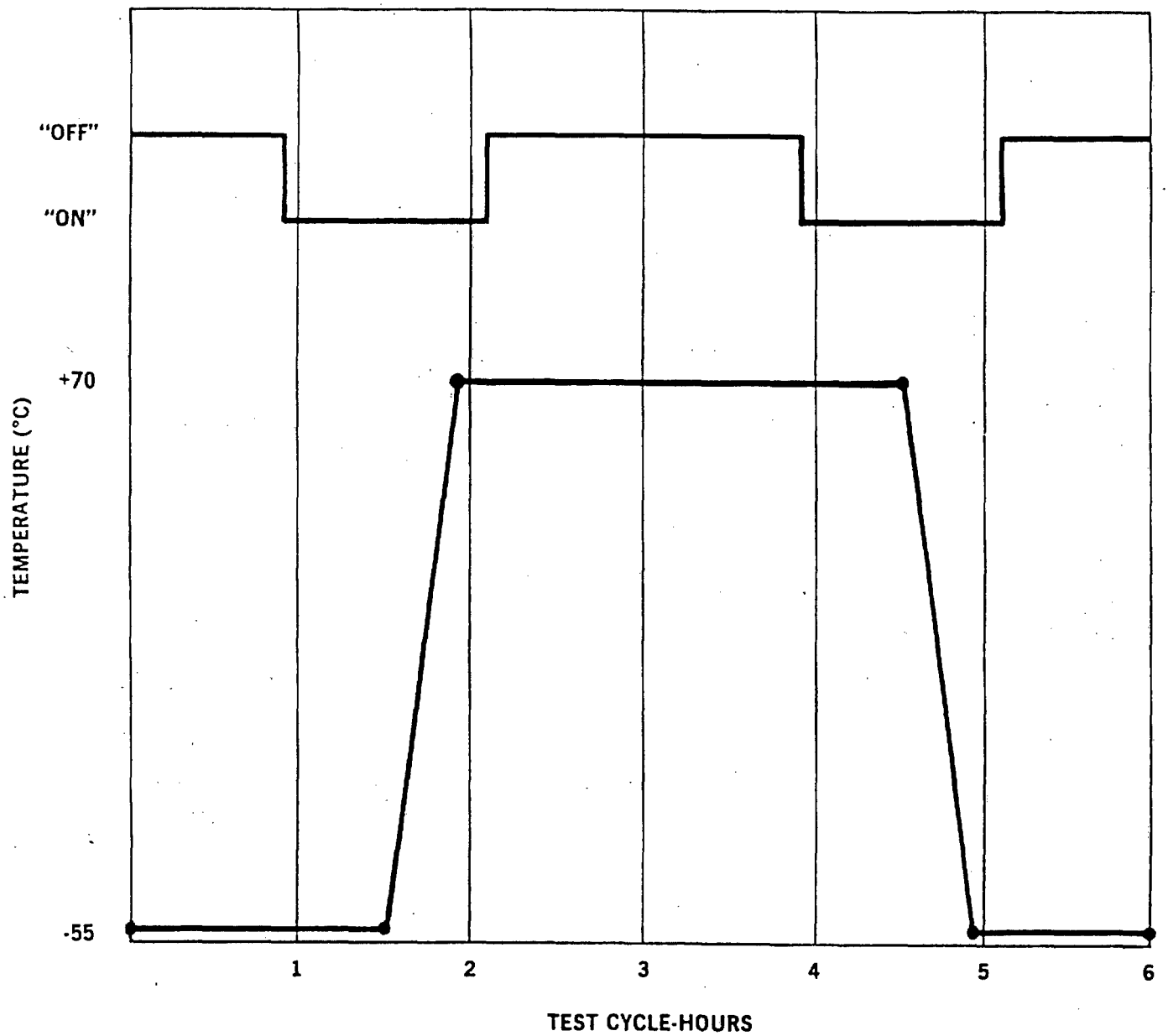
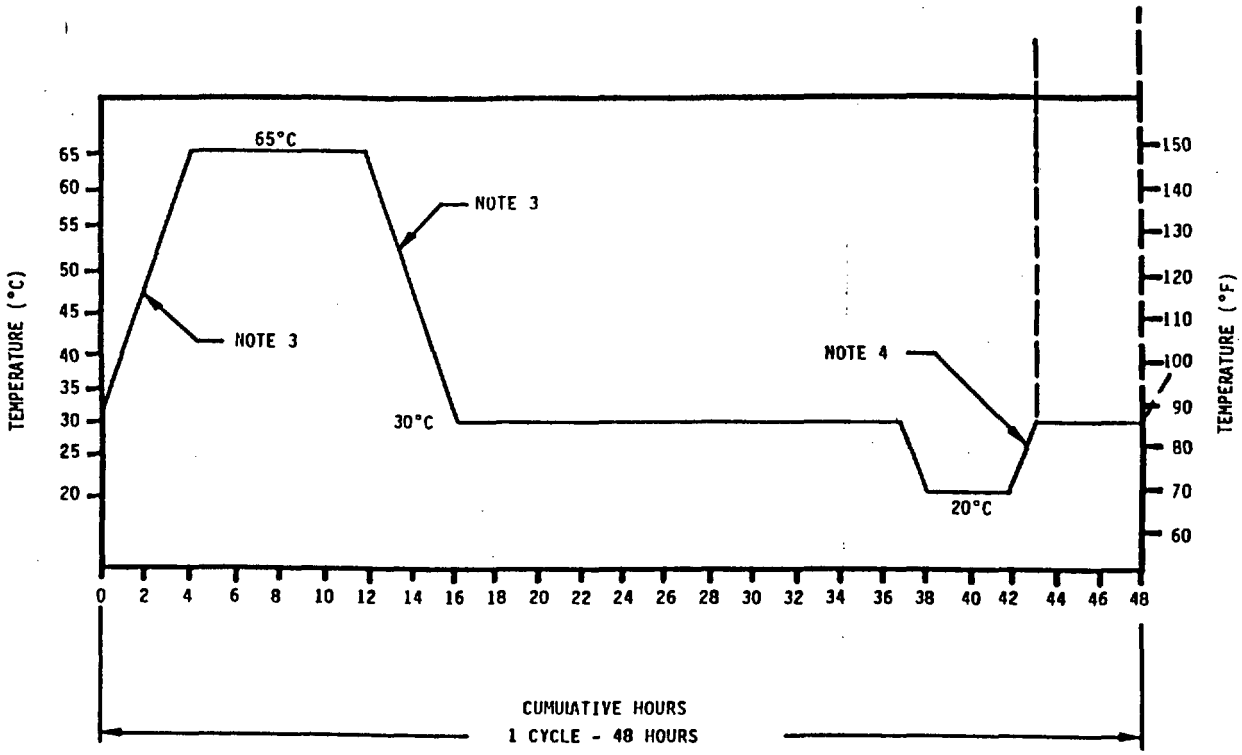


FIGURE 4. Oxygen monitor temperature cycle.

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NOTES:

1. Tolerance during temperature change shall be not greater than 3°C (5°F).
2. Relative humidity shall be maintained at 94 ± 4 percent at all times, except that during the descending temperature period, the relative humidity may be permitted to drop as low as 85 percent.
3. Rate of temperature change between 30° and 65°C (86° and 149°F) shall be not less than 8°C (14.4°F) per hour.
4. The temperature increase in this portion of the curve shall be not less than 10°C (18°F).

FIGURE 5. Humidity cycle.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-85522A(AS)	91/06/17

3. DOCUMENT TITLE
MONITOR, OXYGEN, CRU-91/A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Commanding Officer NAEC, SESD, Code 53	(1) Commercial (908) 323-1280	(2) AUTOVON 624-1280
c. ADDRESS (Include Zip Code) Lakehurst, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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